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Optimising 7T-fMRI for Imaging Regions of Magnetic Susceptibility

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ABSTRACT

The temporal signal-to-noise ratio (tSNR) of functional magnetic resonance imaging (fMRI) is particularly poor in ventral anterior temporal and orbitofrontal regions because of B_0 and B_1^+ magnetic field inhomogeneity, a problem that is exacerbated at higher field strengths. In this 7T-fMRI study we compared three methods of improving sensitivity in these areas: parallel transmit, which uses multiple transmit elements, controlled independently, to homogenise the flip angle experienced by the tissue; multi-echo, which entails collection of multiple volumes at different echo times following a single radiofrequency pulse; and multiband, in which multiple slices are acquired simultaneously. We found that parallel transmit and multi-echo increased the magnitude of the BOLD signal change, but only multi-echo increased BOLD magnitude in areas prone to susceptibility artefacts. Multiband and denoising of multi-echo data with independent components analysis (ICA) both improved precision of GLM fit. Exploratory results suggested that multi-echo and ICA denoising can both benefit multivariate analyses. In conclusion, a multi-echo, multiband sequence improved fMRI quality in areas prone to susceptibility artefacts while maintaining sensitivity across the whole brain. We recommend this approach for studies investigating the functional roles of ventral temporal and orbitofrontal regions with 7T fMRI.

1 | Introduction

The temporal signal-to-noise ratio (tSNR) of functional magnetic resonance imaging (fMRI) varies across the brain. The ventral anterior temporal cortex and orbitofrontal cortex, for example, are affected by two kinds of magnetic field inhomogeneity. First, these regions are located next to air-filled sinuses and so are affected by B_0 inhomogeneity that causes signal dropout and geometric distortions (Devlin et al. 2000; Halai et al. 2014, 2015, 2025; Jezzard and Clare 1999). The effects are more severe at higher field strengths, implying that it may be especially difficult to measure task-related activity in susceptible regions with ultra-high-field fMRI (e.g., 7T-fMRI). Second, at 7T and above, the radiofrequency wavelength is similar to the size of the head, resulting in standing wave effects that cause B_1^+ inhomogeneity in the temporal lobes (Gras et al. 2019; van der Kolk et al. 2013). This in turn causes spatial variation in flip angle and therefore in image intensity (Uğurbil 2018; Wu et al. 2018). B_0 and B_1^+ inhomogeneity make it challenging to use fMRI to investigate the roles that these regions may play in a myriad of cognitive processes—including vision (Devereux et al. 2018), language (Borghesani et al. 2016), multimodal semantic cognition (Lambon Ralph et al. 2017), emotion (Fernandez et al. 2017), social cognition (Binney et al. 2016; Zahn et al. 2007), theory of mind (DuPre et al. 2016) and executive function (Duncan 2010). However, 7T-fMRI also has many advantages in regions unaffected by inhomogeneity: 7T-fMRI offers improved tSNR relative to 3T-fMRI (Morris et al. 2019), which can be used to reduce voxel size and enable applications such as laminar fMRI (Koopmans et al. 2011) or to reduce acquisition times and enable shorter scan times for special populations such as patients with neurodegenerative diseases (Cope et al. 2023). 7T-fMRI also benefits from improved spatial specificity relative to 3T-fMRI because the signal from cortical microvasculature is enhanced while the signal from large veins is reduced (Marques and Norris 2018). Improving signal homogeneity would allow researchers studying the whole brain (or focusing on regions prone to susceptibility artefacts) to take full advantage of 7T-fMRI; therefore, in this study we compared three methods of doing so.

One possible method for improving image quality is parallel transmit (pTx), which uses multiple transmit elements, controlled independently, to homogenise the flip angle pattern experienced by the tissue (Adriany et al. 2005; Deniz et al. 2019; Roemer et al. 1990; Uğurbil 2018; Van de Moortele et al. 2005). In this way, pTx directly counteracts the effect of B_1^+ inhomogeneity; however, since the design of spoke pTx pulses incorporates B_0 field maps, pTx is also capable of counteracting B_0 inhomogeneity to an extent (Zhang et al. 2024). A recent study used pTx to improve image quality in ventral anterior temporal regions for echo-planar imaging (EPI) 7T fMRI (Ding et al. 2022). pTx improved tSNR across the brain compared to a standard sequence, particularly in the temporal lobes. However, there was no improvement in functional contrast during a semantic association task that is known to recruit the anterior temporal lobes in 3T-fMRI studies (Jung et al. 2017).

A second method of recovering signal in these regions is multi-echo (ME) imaging (Kundu et al. 2017; Poser et al. 2006; Posse 2012), which aims to counteract the effects of B_0 inhomogeneity. T_2* is known to vary across the brain (Hagberg et al. 2002); in areas affected by B_0 inhomogeneity, T_2* is particularly short due to increased intravoxel dephasing. A single echo provides sensitivity to a narrow range of T_2* values; the echo time (TE) is therefore selected to provide the best compromise of sensitivity to T_2* across the whole brain. Combining data from multiple echoes increases the range of T_2* that can be imaged with high fidelity. ME has been shown to improve functional contrast (Poser and Norris 2009) and spatial specificity (Boyacioğlu et al. 2015) at 7T and 3T (Fernandez et al. 2017; Halai et al. 2025; Kirilina et al. 2016; Lynch et al. 2020). Having multiple echoes also facilitates the separation of signal and noise because signals decay in a well-characterised way across echoes, whereas noise does not. This principle underpins multi-echo independent components analysis (ME-ICA), via which ICA components that are TE-independent, and thus are likely to be noise rather than blood-oxygen-level-dependent (BOLD) signal, can be removed (Dipasquale et al. 2017; Kundu et al. 2011, 2013, 2015, 2017). This method may enhance signal detection in areas prone to susceptibility artefacts on top of the advantage offered by ME alone (e.g., Lombardo et al. 2016). ME sequences have some potential disadvantages. For example, ME can lengthen repetition time (TR). In-plane acceleration is frequently needed to achieve a sufficiently short first TE, which reduces tSNR (Yun and Shah 2017). In turn, a short first TE, combined with hardware constraints, often limits the minimum voxel size (Koopmans et al. 2011). Critically, in many previous studies examining the benefits of ME sequences compared to single-echo (SE), the 'single-echo' data were extracted from the ME dataset (Amemiya et al. 2019; Bhavsar et al. 2014; Caballero-Gaudes et al. 2019; Cohen et al. 2017, 2018; Dipasquale et al. 2017; Evans et al. 2015; Fernandez et al. 2017; Gilmore et al. 2022; Kovářová et al. 2022). This means that the 'single-echo' data will inherit suboptimal parameters that are ME-specific, making the comparison unfair.

A third strategy for improving acquisition is multiband (MB) imaging, also known as simultaneous multi-slice, in which multiple slices are acquired simultaneously (Barth et al. 2016; Moeller et al. 2010; Setsompop et al. 2012). Rather than counteracting B_0 or B_1^+ inhomogeneity directly, MB has typically been applied to increase temporal resolution—multiple studies have shown that MB can reduce noise aliasing, increase statistical power and counteract the increase in TR associated with ME (Feinberg et al. 2010; Griffanti et al. 2014; Halai et al. 2025; Puckett et al. 2018; Smith et al. 2013). Benefits specific to task fMRI have been less clear (Demetriou et al. 2018; Todd et al. 2016) since some noise features are unlikely to be correlated with task regressors and can be removed with high-pass filtering. Other possible disadvantages of MB include a reduction in tSNR due to increases in g-factor effects (Demetriou et al. 2018; Risk et al. 2021; Setsompop et al. 2012) and leakage of signal into the simultaneously-excited slices (Todd et al. 2016).

Previous work therefore makes clear predictions about which 7T-fMRI sequences are capable of improving sensitivity in regions affected by B_0 and B_1^+ inhomogeneity without compromising signal quality elsewhere. In this study we tested five possible sequences with the goal of demonstrating their *relative* impacts on functional effects of interest. These consisted of a 2×2 factorial design varying number of echoes and multiband factor—single-echo single band (SESB), single-echo multiband (SEMB), multi-echo single band (MESB), and multi-echo multiband (MEMB)—plus a fifth sequence that used pTx. Rather than attempting to map the limits of each parameter, our aim was to generate sequences that would be of practical benefit to neuroscientists seeking to answer a range of cognitive questions that implicate regions affected by inhomogeneity.

We used a semantic judgment task that is known to evoke activity across the semantic network, including areas severely affected and those relatively unaffected by susceptibility artefacts (Jung et al. 2017).

We had two univariate effects of interest—activation magnitude and activation precision (Halai et al. 2025)—and we hypothesised that our different sequences, which each aim to improve signal quality in different ways, would affect these metrics differently. Activation magnitude is the magnitude of the BOLD signal change, operationalised as the first-level beta values extracted from each voxel. We hypothesised that both pTx and ME would recover signal and hence increase activation magnitude relative to the SESB sequence (which we used as a baseline). Activation precision is the reliability of the BOLD signal change, analogous to the functional contrast-to-noise ratio (fCNR) and operationalised as the first-level *t* values extracted from each voxel. We hypothesised that MB sequences, with greater effective degrees of freedom, would increase activation precision relative to single band (SB) sequences. We also had two

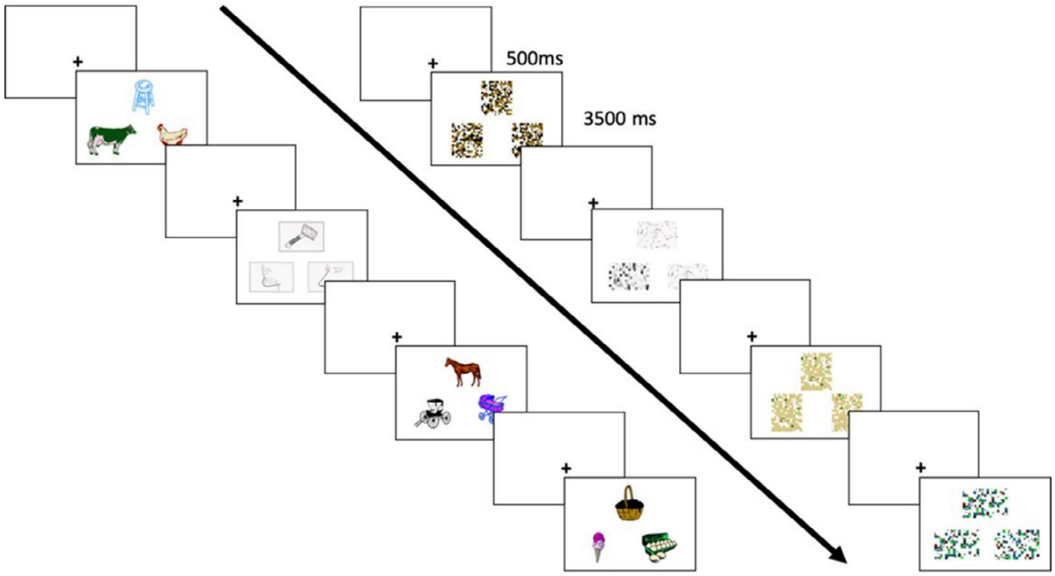

FIGURE 1 | One block of the semantic association task (left of the arrow) and one block of the control task (right of the arrow). Each trial consisted of a fixation cross (500 ms) followed by three pictures presented simultaneously. Participants had to select which of the two pictures at the bottom was more semantically similar to (in the semantic task) or visually identical to (in the control task) the picture at the top (the probe picture). Each block lasted 16 s in total. *Source:* Reprinted with permission from Halai et al. (2025).

secondary hypotheses: first, since ME-ICA denoising removes TE-independent noise, greater activation precision would be observed in multi-echo denoised data (MEdn) relative to ME data without denoising; second, that any MB advantage would be due to the increase in the number of volumes.

Our study focused primarily on univariate effects. However, multivariate analysis techniques, which exploit variance and covariance between voxels and can accommodate participant-specific differences in activation patterns (Coutanche 2013; Davis et al. 2014; Davis and Poldrack 2013), are rapidly gaining popularity. These methods frequently rely on the assumption of good-quality signal across the whole brain (Frisby et al. 2023), and so we conducted an exploratory analysis, following the method of Haxby et al. (2001), to decode task condition from the data acquired with each sequence. We hypothesised that pTx and ME sequences could improve MVPA performance by recovering signal, that MB could improve MVPA performance by increasing the effective degrees of freedom, and that ME-ICA could improve MVPA performance by removing TE-independent noise, so we tested all of these comparisons. Finally, we tested for the presence of slice leakage artefacts in our MB data.

To summarise, our study aimed to compare pTx, ME and MB as methods for improving sensitivity in ventral temporal and orbitofrontal regions while maintaining image quality across the rest of the brain.

2 | Methods

2.1 | Participants

Twenty healthy native speakers of British English (age range 18–50, mean age 33.45 years, 12 female, 8 male) participated in the study. All were right-handed, had normal or corrected-to-normal vision, and had no neurological or sensory disorders. All participants gave written informed consent. The research was approved by a local National Health Service (NHS) ethics committee (04/Q105/66).

2.2 | Stimuli and Task

All participants performed a semantic association task and a visual pattern matching task (hereafter called the 'control task') adapted from a previous study (Jung et al. 2017). Each stimulus consisted of three pictures presented simultaneously (Figure 1). Some pictures were line drawings taken from the Pyramids and Palm Trees test (Howard and Patterson 1992) and some were colour cartoons or photographs taken from the Camel and Cactus test (Bozeat et al. 2000). In the semantic task, participants were instructed to indicate which of the two pictures at the bottom of the screen had the closest semantic relationship to the picture at the top (hereafter the 'probe picture'). In the control task, participants were instructed to indicate which of two scrambled pictures (generated from the pictures used in the semantic task) at the bottom of the screen was identical to a scrambled probe. There were 248 unique picture triplets, so some stimuli were repeated between runs, but no stimulus was repeated within a run. E-Prime software (Psychology Software Tools Inc., Pittsburgh, USA) was used to display stimuli and record responses. Stimuli were rear-projected onto a screen at the back of the MRI scanner, and observers viewed stimuli through a mirror mounted to the head coil directly above the eyes.

The study had a block design with three types of block: semantic, control and rest. Each task block consisted of four trials. Each trial consisted of a fixation cross presented for 500 ms followed by a stimulus presented for 3500 ms. Each rest block consisted of a fixation cross presented for 16 s. Each run began 16 s after the start of the MR sequence and then consisted of 30 blocks

TABLE 1 | Parameters of each sequence.

	SESB	SEMB	MESB	MEMB	pTx
No. echoes	1	1	3	3	1
Multiband factor	1	2	1	2	1
TR (ms)	3020	1510	3020	1510	3000
TE_1 (ms)	25.00	25.00	11.80	11.80	25.00
TE_2 (ms)	—	—	27.05	27.05	—
TE_3 (ms)	—	—	42.30	42.30	—
iPAT type	Off	Off	GRAPPA	GRAPPA	GRAPPA
iPAT factor	Off	Off	3	3	2
Phase partial Fourier	7/8	7/8	7/8	7/8	Off
Nominal flip angle (°)	50	50	78	63	40
Bandwidth (Hz/Px)	2204	2204	2204	2204	1804
Number of volumes acquired	171	340	171	340	172
Number of pulses	1	1	1	1	5 (VERSE 2-spoke pTx pulses; duration 5.12 ms)

presented in the order: semantic, control, semantic, control, rest. There were five runs per participant, collected in a single session. Accuracy and reaction time were measured for each trial. Since reaction time is not normally distributed, both metrics were compared across tasks using Wilcoxon's signed-rank tests and across sequences using Friedman's nonparametric ANOVA.

2.3 | Image Acquisition

All images were acquired on a whole-body MAGNETOM Terra 7T MRI (Siemens Healthcare, Germany). An 8Tx32Rx head coil (Nova Medical, USA) was used to run all structural and functional imaging sequences.

An MP2RAGE anatomical scan was acquired with the following parameters: 224 sagittal slices (interleaved acquisition), FOV $240 \times 225.12 \times 240\,mm^3$, voxel size 0.75 mm isotropic, TR 4300 ms, inversion times 840 and 2370 ms, TE 1.99 ms, nominal flip angles 5° and 6°, GRAPPA acceleration factor 3, and duration 8 min 50 s.

Next, manual B_0-shimming was performed over the volume to be used for EPI. This was necessary because of a known bug in the scanner software which rendered automatic shims inaccurate. A double-echo steady state (DESS) sequence was used to generate a field map and the radiographer adjusted shim parameters manually with the aim of reducing the water linewidth, defined as full width of the spectrum at half height, to below 40 Hz. However, for some participants the adjustments proved time-consuming and so adjustment time was capped at 30 min from the start of scanning after which the best shim parameters were adopted. The actual mean water linewidth over participants was 42 Hz (standard deviation over participants = 9 Hz; missing data for four participants). Then a dummy pTx-EPI scan was acquired to trigger the acquisition of subject-specific B_0 and per-channel B_1^+ field maps. Parameters of this dummy scan were the same as the pTx functional sequence (see Table 1 and below) except that only one volume was acquired. The brain was divided into five slabs along the slice direction and slab-specific 2-spoke pTx excitation pulses were designed offline. Variable-rate selective excitation (VERSE; Hargreaves et al. 2004) was applied to reduce specific absorption rate (SAR) for the pTx sequence only.

There were five functional runs of EPI, one run of each sequence. Key parameters are given in Table 1. To ensure that our results would be of use to future cognitive neuroimaging studies, the sequences were designed to represent the way that each parameter would be used in a typical study. Three echoes were used because this is the minimum number needed for ME-ICA and because our scanner hardware prevented us from using four (Halai et al. 2025). An MB factor of 2 was used because previous work suggested that this MB factor, combined with a GRAPPA factor of 3, would be unlikely to cause slice leakage (Todd et al. 2016). GRAPPA and phase partial Fourier were used to ensure a reasonable TE and TR for ME sequences and to ensure that the SE and pTx sequences would have comparable TRs. However, in-plane acceleration reduces tSNR (Yun and Shah 2017) and using *both* GRAPPA and phase partial Fourier is not needed to achieve a reasonable TE for a single-echo sequence. To facilitate a fair comparison of ME and SE sequences as they would be used in typical studies, no GRAPPA was used for SE sequences. Two spokes were used because, as the number of spokes increases, the pulse length increases and the bandwidth decreases, and using two spokes keeps both those parameters within a reasonable range. Five slabs were used because, as the number of slabs increases, the time needed for pulse calculation also increases, and using five slabs keeps the calculation time short (~5 min; Ding et al. 2022). Due to experimenter error, the flip angle was not set to the Ernst angle for SE sequences. The possible impact of this error on results is explored in Supplementary Material 2. Bandwidth of the pTx sequence was selected to match previous work (Ding et al. 2022). It was necessary to increase the bandwidth for our other sequences in order to control TE, but doing so reduces tSNR. To facilitate a fair comparison of pTx to the other sequences as they would be used in typical studies, we did not lower the bandwidth of the pTx sequence to match the others.

The order of sequences was counterbalanced across participants. The following parameters were held constant across sequences: 48 axial slices (interleaved acquisition), FOV $210 \times 210 \times 210\,\text{mm}^3$ to cover the whole brain in most participants (visual inspection ensured that the ventral anterior temporal lobe was included in the FOV for all participants and the FOV was tilted up at the nose to avoid ghosting of the eyes into the temporal lobe), voxel size 2.5 mm isotropic (no gap), and A-P phase encoding direction. After each run, five further volumes were acquired with the phase encoding direction changed to P–A to facilitate distortion correction during preprocessing.

2.4 | Data Analysis

All analysis code is available at https://github.com/slfrisby/7TOptimisation/.

2.4.1 | Preprocessing

For ease of data sharing we converted all DICOMs to BIDS format (Gorgolewski et al. 2016) using *heudiconv* v1.0.0 (Halchenko et al. 2024).

Standard reproducible preprocessing pipelines designed for 3T-fMRI, such as *fMRIprep* (Esteban et al. 2019; Gorgolewski et al. 2011; Markiewicz et al. 2024) perform poorly on 7T-fMRI EPI data. Therefore, the analysis pipeline was split into stages using different software packages.

Since it is notoriously difficult to perform a good-quality brain extraction on MP2RAGE data (because of salt-and-pepper noise in the background and cavities), two pipelines were used. The MP2RAGE T1w (combined) image first had its background noise removed using O'Brien regularisation (O'Brien et al. 2014) and was then submitted to the CAT12 pipeline for segmentation (Gaser et al. 2024; in SPM12; https://www.fil.ion.ucl.ac.uk/spm/). The bias- and global-intensity corrected T1w image produced was provided to as input to the anatomy pipeline in *fMRIPrep* 21.0.1. TheT1w image was skull-stripped with a Nipype implementation of the *antsBrainExtraction.sh* workflow (ANTs 2.3.3; Avants et al. 2009, 2011; https://github.com/ANTsX/ANTs/). Volume-based spatial normalisation to standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with *antsRegistration.sh* (ANTs).

Functional preprocessing was performed using in-house code, composed of functions from AFNI (v.18.3.03; Cox 1996; Cox and Hyde 1997; https://afni.nimh.nih.gov/), FSL (v.5.0; Andersson et al. 2003; Jenkinson et al. 2012; Smith 2002; Smith et al. 2004; https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/), tedana (v.23.0.1; DuPre et al. 2021; Kundu et al. 2011, 2013; The tedana Community et al. 2023; https://tedana.readthedocs.io/en/stable/index.html) and ANTS (v. 2.2.0; Avants et al. 2009, 2011; https://github.com/ANTsX/ANTs/). EPIs were despiked using *3dDespike* (AFNI); slice timing was corrected to the middle slice using *3dTshift* (AFNI); motion was corrected with *3dvolreg* and *3dAllineate* (AFNI) using the first volume of each run as a reference (for ME datasets, the TE_1 image was aligned and the resulting transform was applied to the TE_2 and TE_3 images); and skull-stripped using *BET* (FSL) to create a participant-specific brain mask.

For ME datasets only, *tedana* was used to create two timeseries—one in which the echoes were optimally combined based on T_2* weighting (Posse 2012), and one in which the T_2* optimally-combined data were denoised using ICA. *tedana* conducts denoising by decomposing data using PCA and ICA, classifying components according to whether the signal scales linearly with TE (as the BOLD signal does), and reconstructing the data using only BOLD-like components. The brain mask created with *BET* was used as the mask for this stage.

Next, for all datasets, unwarping was conducted using *topup* and *applytopup* (FSL). Field displacement maps were calculated using 10 volumes (five with A–P phase encoding direction, extracted from the start of each functional run, and five with P–A phase encoding direction, collected separately after each run) and the resulting correction was applied to all images. Finally, the mean EPI for each run was coregistered to the skull-stripped native structural image using a rigid-body registration with *AntsRegistrationSyN.sh* (ANTS). EPIs were then transformed into standard space (MNI152NLin2009cAsym) by combining the transforms from native EPI to native T1 and the transforms from native T1 to standard space and applying those transforms to the EPIs using *antsApplyTransforms* (ANTS). Images were smoothed with a 6 mm FWHM Gaussian filter in SPM12 (https://www.fil.ion.ucl.ac.uk/spm/) for GLM analysis.

A separate functional preprocessing pipeline was used to create images for slice leakage analysis. This pipeline differed from the main pipeline in the following ways. Despiking was omitted to avoid the removal of noise. For ME datasets only, rather than run the full *tedana* workflow, we conducted optimal combination of data from multiple echoes (but no denoising) using the *t2smap* command using all voxels in the volume (i.e., no brain mask). All coregistration steps were omitted (the images remained in native EPI space) and no smoothing was applied.

2.4.2 | First-Level (Within-Participant) GLM

Data were analysed using the general linear model (GLM) approach implemented in SPM12 in MATLAB r2019a. We had five timeseries of primary interest—standard single-echo single band (SESB), parallel transmit (pTx), single-echo multiband (SEMB), multi-echo single band (MESB) and multi-echo multiband (MEMB). For the latter two sequences, data from all echoes were optimally combined but were not ICA-denoised. We also generated two timeseries with ME-ICA denoising—multi-echo single band with ICA denoising (MESBdn) and multi-echo multiband with ICA denoising (MEMBdn)—and two downsampled MB timeseries created by extracting odd-numbered volumes to match the number of volumes in the single band timeseries—odd-numbered volumes of single-echo multiband (SEMBodd) and odd volumes of multi-echo multiband (MEMBodd).

At the individual subject level, each block of the semantic and control task was modelled as a boxcar function (resting blocks were modelled implicitly) and these boxcar functions were subsequently convolved with SPM's difference of gammas haemodynamic response function. The six motion parameters extracted during preprocessing were used as regressors of no interest. The micro-time resolution was set as the number of slices ($n = 48$), the micro-time onset was set as the reference slice for slice-time correction ($n = 24$), and the high-pass filter cutoff was 128 s. The same MNI template used during preprocessing was used as a mask for the analysis. The parameter estimation method was restricted maximum likelihood estimation (ReML), and serial correlations were accounted for using an autoregressive AR(1) model during estimation. For univariate analyses, the contrast of interest was greater activation for the semantic task than the control task (S > C). For exploratory multivariate pattern analysis (MVPA), each block (12 semantic and 12 control) was modelled individually in order to obtain one beta image per block.

Finally, for the slice leakage analysis, the modelling was rerun on the minimally-preprocessed timeseries without any brain mask. We obtained both univariate contrasts of interest (S > C) and beta images per block (for MVPA).

2.4.3 | Second-Level (Across-Participant) GLM

2.4.3.1 | Region-of-Interest (ROI) Analysis. Regions of interest (ROIs) were defined based on a large-scale distortion-corrected 3T-fMRI study of the semantic network (Humphreys et al. 2015). For each comparison, ROIs were analysed only if they overlapped by at least one voxel with the whole-brain contrast of interest (S > C) summed over all sequences in the comparison.

2.4.3.1.1 | Univariate Analysis. Activation magnitude (first-level beta values) and activation precision (first-level t values) were extracted from each ROI using a publicly-available script, *roi_extract.m* (https://github.com/MRC-CBU/riksneurotools/blob/master/Util/roi_extract.m). There were two planned t-tests for activation magnitude (pTx > SESB, ME > SE) and four planned t-tests for activation precision (MB > SB, MEdn > ME, MBodd > SB, SB > MBodd). For each planned t-test, results were Bonferroni-corrected for the number of ROIs included.

2.4.3.1.2 | Exploratory Multivariate Pattern Analysis (MVPA). The input to this analysis was beta images, one image per block (12 semantic and 12 control). For each block and each ROI, a vector v of beta values was created (voxels within the ROI are numbered arbitrarily):

$$v_{\text{block } n} = \left[\beta_{\text{voxel } 1}, \beta_{\text{voxel } 2}, \dots \beta_{\text{voxel } n}\right]$$

For each ROI, the cosine distance D (1—cosine similarity; Halai et al. 2025) between every possible pair of blocks was calculated. For example:

$$D_{\text{block } 1,\ \text{block } 2} = 1 - \left(\frac{v_{\text{block } 1} \cdot v_{\text{block } 2}}{\| v_{\text{block } 1} \| \ \| v_{\text{block } 2} \|} \right)$$

For each ROI, MVPA performance was operationalised as the mean between-task distance (over every possible pair of blocks of different tasks) minus the mean within-task distance (over every possible pair of blocks of the same task). Where block S is a block of the semantic task and block C is a block of the control task, and n_{blocks} is the total number of blocks of each task (12):

$$\text{MVPA metric} = \left(\frac{\sum D_{\text{block } S1,C1}, D_{\text{block } S1,C2}, \dots, D_{\text{block } Sn,\text{block } Cn}}{n_{\text{blocks}}^{2}} \right) - \left(\frac{\sum D_{\text{block } S1,\text{block } S2}, \dots, D_{\text{block } Sn-1,\text{block } Sn}, D_{\text{block } C1,\text{block } C2}, \dots,\ D_{\text{block } Cn-1,\text{block } Cn}}{2\left(n_{\text{blocks}}^{2}\right)} \right)$$

If a region is encoding information about the task, we would expect mean within-task similarity to be large and mean between-task similarity to be small, so the MVPA metric would take a large negative value. If a region is not encoding information about the task, we would expect both within-task and between-task similarity to be zero and the difference between them to be negligible (Halai et al. 2025).

Paired t-tests were used to compare the MVPA metric across sequences (all planned t-tests described in the univariate ROI analysis were conducted; Haxby et al. 2001).

2.4.3.2 | Whole-Brain Analysis. All the above contrasts were assessed at the whole-brain level using t-tests (for comparing pTx and SESB) or using random-effects ANOVAs with one-sample t-tests on the summary statistic (for the three factorial designs: varying echo and band (SESB, SEMB, MESB and MEMB); varying denoising and band (MESBdn, MEMBdn, MESB and MEMB); and varying echo and downsampled band (SEMBodd, MEMBodd, SEMB and MEMB)). The ANOVAs were conducted using a publicly-available script, *batch_spm_anova.m* (https://github.com/MRC-CBU/riksneurotools/blob/master/SPM/batch_spm_anova.m). The group t-maps were assessed for significance by using a voxel-height threshold of $p < 0.001$ to define clusters and then a cluster-defining family-wise-error corrected threshold of $p < 0.05$ for statistical inference.

2.4.3.3 | Slice Leakage Analysis. The group-level, whole-brain contrast of interest (S > C) in standard space was inspected and the coordinates of the peak t-values of the top five clusters were identified. These coordinates were back-projected to obtain five sets of corresponding coordinates in each participant's native EPI space (hereafter 'seeds', labelled A). Next, voxels to which signal might be warped were identified as artefact locations based on phase shift (FOV/2, labelled B) and, in the MEMB data, GRAPPA (labelled Ag) and phase shift plus GRAPPA (labelled Bg). A spherical ROI, four voxels in radius, was defined around

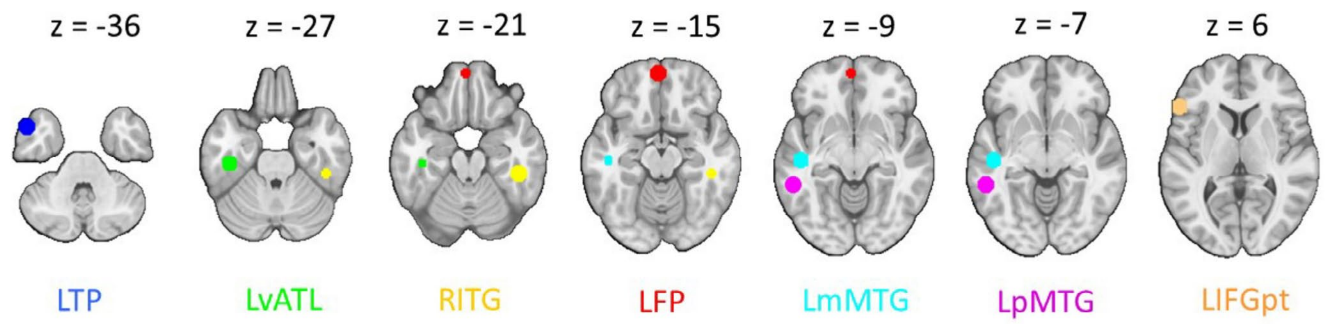

FIGURE 2 | Regions of interest, taken from a meta-analysis of semantic tasks by Humphreys et al. (2015). All spheres are 8 mm in radius. Regions of interest are overlaid on the MNI152NLin2009cAsym template. LFP = left frontal pole; LIFGpt = left inferior temporal gyrus pars triangularis; LmMTG = left medial middle temporal gyrus; LpMTG = left posterior middle temporal gyrus; LTP = left temporal pole; LvATL = left ventral anterior temporal lobe; RITG = right inferior temporal gyrus.

each seed location and possible artefact location using a modified version of the scripts developed for McNabb et al. (2020, https://github.com/DrMichaelLindner/MAP4SL/, our version available at https://github.com/slfrisby/7TOptimisation/).

Both univariate (McNabb et al. 2020) and multivariate (Halai et al. 2025) slice leakage tests were conducted. Activation magnitude was extracted from minimally-preprocessed data and for the multivariate analysis, the difference between mean within-task similarity and mean between-task similarity was calculated as in the ROI analysis. Paired t-tests were conducted for each seed and artefact location between each sequence of interest (SEMB and MEMB) and a control sequence. For artefact locations based on phase shift (B), the control sequence was the corresponding SB sequence (SESB for SEMB and MESB for MEMB). For artefact locations based on GRAPPA, the control sequence was the corresponding SE sequence because SE sequences were collected without GRAPPA (SEMB for MEMB). Results were Bonferroni-corrected for the number of peaks ($n = 5$) and the number of possible artefact regions ($n = 1$ for SEMB and $n = 3$ for MEMB).

3 | Results

3.1 | Excluded Participants

Two participants were excluded because of excessive head motion (this was defined by calculating, for each participant, the percentage of volumes per run with absolute translation values of over 2 mm or absolute rotation values over 1°, averaging these percentages over runs, and excluding any participant whose mean percentage was greater than two standard deviations above the mean percentage across participants). One participant was excluded because of technical problems during data acquisition which meant that the pTx run failed. All subsequent analyses were conducted on the remaining 17 participants.

3.2 | Behavioural Results

The 17 participants had good performance on both tasks and, importantly, there were no significant differences between the five sequences in terms of accuracy (Friedman's $\chi^2 = 5.26$, $p = 0.26$) or reaction time (Friedman's $\chi^2 = 7.20$, $p = 0.16$). The semantic and control tasks did not differ reliably in accuracy (Wilcoxon's $z = 0$; $p = 0.06$) or reaction time (Wilcoxon's $z = 7$; $p = 1.00$).

3.3 | Region-of-Interest Analysis

ROIs that overlapped with the whole-brain contrast of interest (S > C) for at least one of the five sequences are shown in Figure 2.

3.3.1 | Univariate Analysis

Table 2 shows the p values for all planned t-contrasts. pTx provided significantly better activation magnitude than SESB in the left posterior middle temporal gyrus (LpMTG). ME sequences provided significantly better activation magnitude than SE sequences in the left ventral anterior temporal lobe (LvATL).

MB sequences offered significantly better activation precision than SB sequences in the LvATL, right inferior temporal gyrus (RITG) and left inferior frontal gyrus pars triangularis (LIFGpt). MEdn sequences offered significantly better activation precision than ME in the LvATL, RITG, LpMTG, LIFGpt and left frontal pole (LFP). There was no significant difference in either direction between MBodd sequences and SB sequences (see Table S1 for detailed results, including effects of every sequence on activation magnitude and activation precision plus reverse contrasts). We also found no significant interaction between ME and MB.

3.3.2 | Exploratory MVPA

Table 2 also shows the p values for t-tests comparing our MVPA metric between sequences. The ME sequences produced significantly better performance than SE sequences in the LvATL and LpMTG. MEdn sequences produced better performance than ME sequences in every ROI. All other comparisons failed to reach significance.

3.4 | Whole-Brain Analysis

Figure 3 shows the results for activation magnitude. Figure 3A shows a single cluster, within the right fusiform gyrus and

TABLE 2 | p values for all t-tests within regions of interest (ROIs) based on the semantic network.

	LTP	LvATL	RITG	LFP	LmMTG	LpMTG	LIFGpt
Activation magnitude							
pTx > SESB	—	0.2488	0.0240*	—	—	0.0042**	0.4114
ME > SE	0.0502	0.0001**	0.0333*	0.2951	0.0092*	0.0003*	0.4155
Activation precision							
MB > SB	0.4569	0.0007**	0.0019**	0.0989	0.2187	0.0266*	0.0016**
MEdn > ME	0.3117	< 0.0001**	< 0.0001**	0.0014**	0.1363	< 0.0001**	< 0.0001**
MBodd > SB	0.4225	0.1057	0.0481*	—	0.5697	0.1879	0.3300
MVPA							
pTx > SESB	—	0.8569	0.3280	—	—	0.1529	0.9179
ME > SE	0.0096*	0.0063**	0.1612	0.0655	0.0465*	0.0024**	0.2508
MB > SB	0.2455	0.1417	0.0667	0.3021	0.3949	0.2693	0.5207
MEdn > ME	0.0027**	< 0.0001**	< 0.0001**	0.0023**	0.0001**	0.0012**	0.0007**
SB > MB	0.7416	0.4570	0.7222	—	0.3803	0.6716	0.3242
MBodd > SB	0.2584	0.5430	0.2778	—	0.6197	0.3284	0.6758

Abbreviations: LFP = left frontal pole; LIFGpt = left inferior temporal gyrus pars triangularis; LmMTG = left medial middle temporal gyrus; LpMTG = left posterior middle temporal gyrus; LTP = left temporal pole; LvATL = left ventral anterior temporal lobe; RITG = right inferior temporal gyrus.
* = $p < 0.05$, ** = $p < 0.05$ (Bonferroni-corrected for the number of ROIs), — = ROI does not overlap by at least one voxel with the whole-brain contrast of interest (S > C) summed over all sequences in the comparison.

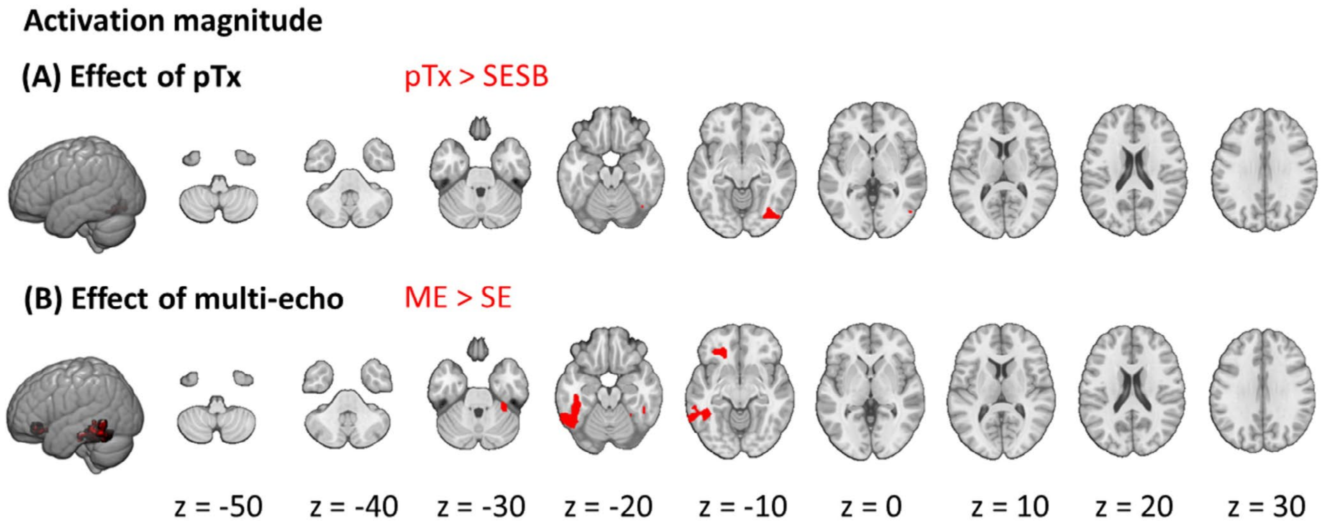

FIGURE 3 | Effects on activation magnitude: (A) effect of parallel transmit (pTx > SESB); (B) effect of echo (ME > SE). Results are cluster-corrected at $p < 0.05$ based on an uncorrected voxel threshold of $p < 0.001$ and are overlaid on the MNI152NLin2009cAsym template.

lateral inferior occipital cortex, that showed greater activation magnitude with pTx than with SESB (pTx > SESB). Figure 3B shows the main effect of ME over SE, featuring clusters in the fusiform and inferior temporal gyri bilaterally plus the left orbitofrontal cortex (cluster and peak information is provided in Table S2).

Figure 4 shows the results for activation precision. Both the main effect of MB over SB and the main effect of ME-ICA denoising over ME without ICA denoising extended down the temporal lobe and included frontal regions (Table S2). There was no significant difference between downsampled MB sequences and SB sequences in either direction.

To summarise, these results aligned with results from our ROI analyses—ME increased activation magnitude and both MB and ME-ICA denoising increased activation precision (see Figures S1–S6 for detailed results, including effects of every sequence on activation magnitude and activation precision plus reverse contrasts).

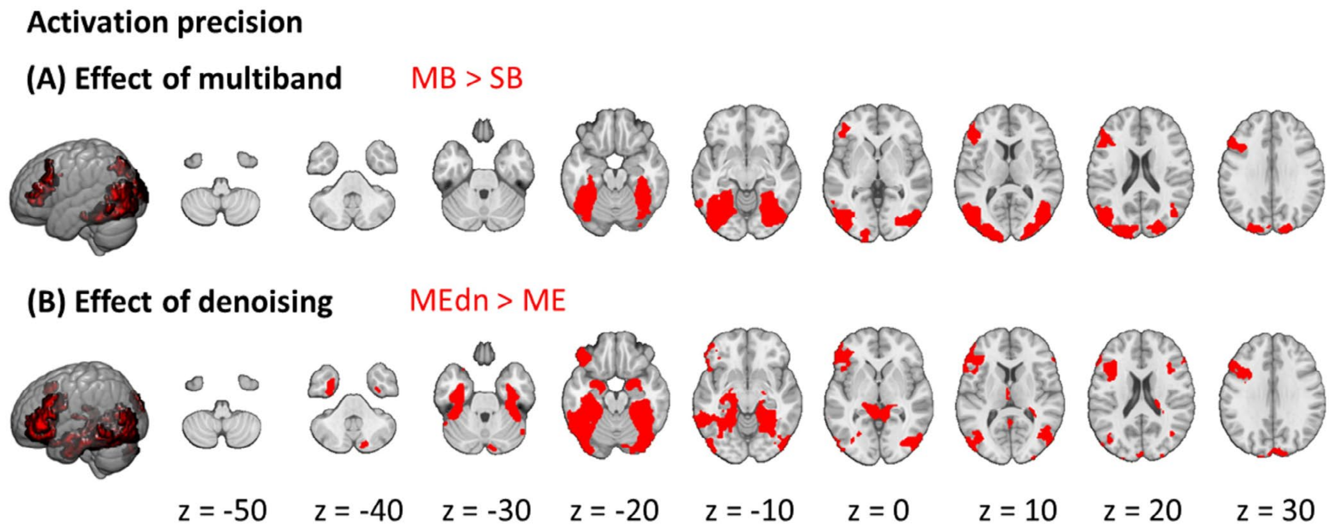

FIGURE 4 | Effects on activation precision: (A) effect of multiband (MB > SB); (B) effect of ME-ICA denoising (MEdn > ME). Results are cluster-corrected at $p < 0.05$ based on an uncorrected voxel threshold of $p < 0.001$ and are overlaid on the MNI152NLin2009cAsym template.

3.5 | Slice Leakage Analysis

Seed and possible artefact locations for an example participant are shown in Figure 5A. Figure 5B shows activation magnitude for possible artefact locations associated with one peak for all participants in each MB sequence and its corresponding sequences. Figure 5C shows MVPA performance. Violin plots for all other peaks are shown in Figure S7. *t*-tests were conducted at each artefact location, but no comparison reached statistical significance (*p* values are shown in Table S3). *t*-tests were not conducted at seed locations—if a difference were discovered, this would be attributable to the MB sequence rather than to leakage of signal from the seed into a different location.

We therefore concluded that there was no evidence for slice leakage in either of our MB datasets.

4 | Discussion

The tSNR of fMRI varies across the brain. This is especially evident in ventral anterior temporal and orbitofrontal regions, which are located next to the air-filled sinuses and are therefore affected by both B_0 and B_1^+ magnetic field inhomogeneity (Devlin et al. 2000; Gras et al. 2019; Halai et al. 2014, 2015, 2025; Uğurbil 2018; van der Kolk et al. 2013; Wu et al. 2018). This study is the first 7T-fMRI study systematically comparing pTx, ME and MB as practical methods for improving sensitivity in these regions while maintaining sensitivity across the brain. We found that pTx improved activation magnitude in posterior temporal and occipital regions. ME, however, resulted in improved activation magnitude extending down the temporal lobe and including inferior frontal regions. Both MB and ME-ICA denoising resulted in improved activation precision in the same areas. In an exploratory analysis we found that ME and ME-ICA improved MVPA performance but MB did not. No slice leakage artefacts were associated with our MB sequences.

Although pTx produced better activation magnitude in posterior temporal and occipital regions than the baseline (SESB) sequence without compromising activation magnitude elsewhere, activation magnitude in anterior temporal areas was comparable for both sequences. These results replicated those of Ding et al. (2022) who failed to find improved task contrast in anterior temporal regions with pTx in spite of improved tSNR in the resting state. Note that, despite visible effects of magnetic field inhomogeneity on the EPIs, even the baseline sequence was able to identify semantic activity extending into ventral anterior temporal regions. This is shown in Figure 6 in which the contour of the whole-brain contrast of interest is overlaid on the mean EPI across all participants (detailed contrast maps are shown in Figures S1 and S3). It is possible that this finding reflects the increased sensitivity of 7T-fMRI compared to 3T-fMRI, where semantic activity is rarely observed without using a method such as ME (Halai et al. 2014, 2015, 2025) or spin-echo (Binney et al. 2010; Embleton et al. 2010). That a standard sequence can detect signal in these regions might come as a surprise to many neuroimaging researchers as 7T-fMRI is not only associated with increased sensitivity, but also with exacerbated magnetic susceptibility artefacts. This finding should therefore embolden researchers to use 7T fMRI for experimental designs that are difficult to conduct at a lower field strength—for example, studies of special populations, who may benefit from shorter scan times (Cope et al. 2023), or studies requiring high spatial specificity (Marques and Norris 2018). We do acknowledge that the way we have utilised 7T fMRI in this study will not be suitable for, for example, studies that require ultra-high resolution (in our case, scanner hardware constrained voxel size when combined with a short first echo time).

The factorial design enabled us to disentangle the effects of ME and MB. As hypothesised, ME improved activation magnitude and these effects were localised to inferior temporal and orbitofrontal areas. Adverse effects on activation magnitude were negligible—although ME reduced activation precision in two very small clusters (Figure S4), these were not within areas that the semantic task is known to recruit (Jung et al. 2017). These results demonstrated that having one very short echo is a successful method of increasing activation magnitude in areas where T_2* is particularly short (Halai et al. 2014, 2015, 2025; Jung et al. 2017). ME also opens up the opportunity for sophisticated denoising such as *tedana* (DuPre et al. 2021; Kundu et al. 2011, 2013; The tedana Community et al. 2023), which improved activation precision in agreement with previous findings (Amemiya et al. 2019; Gonzalez-Castillo et al. 2016). Small clusters in which ME-ICA increased activation magnitude (Figure S5) or decreased activation precision (Figure S6) were not within regions recruited by the task (Jung et al. 2017).

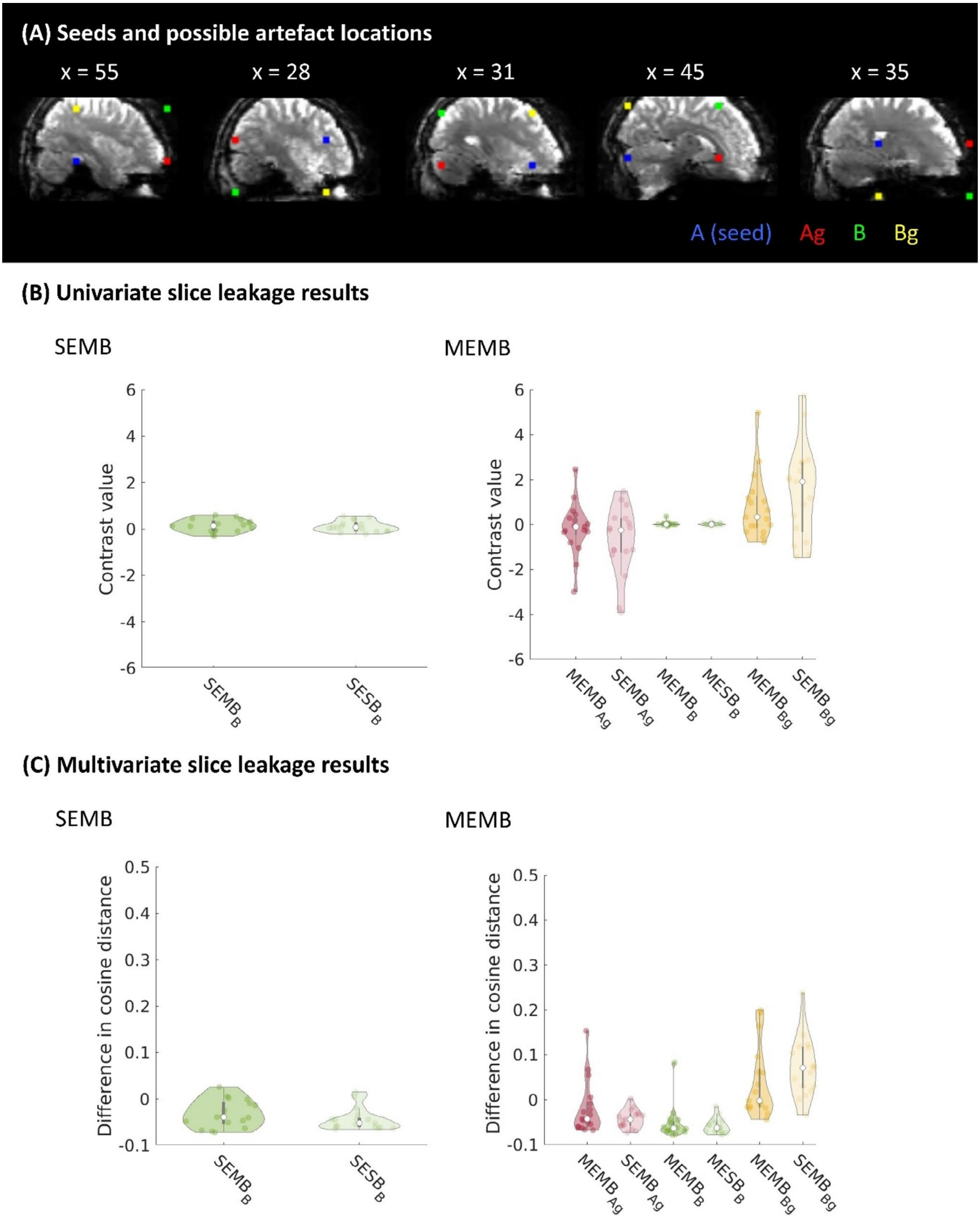

FIGURE 5 | Legend on next page.

FIGURE 5 | Slice leakage analysis. (A) Seed and possible artefact locations for a single participant. Informed consent was obtained from the participant for this image to be published. The ROIs (radius four voxels) indicate the seed location (blue), possible artefact location based on phase shift (green), possible artefact location based on GRAPPA (red), and possible artefact location based on phase shift and GRAPPA (yellow) in native EPI space. Informed consent was obtained from the participant for this image to be published. (B) Group mean activation magnitude within each possible artefact location corresponding to the first seed for each MB sequence and the corresponding control sequences (plots for other seeds are shown in Figure S7; statistics for all seeds are shown in Table S3). Each point within a violin plot indicates the mean for one participant ($n = 17$). (C) Group mean MVPA distance within each sphere for each MB sequence and the corresponding control sequences.

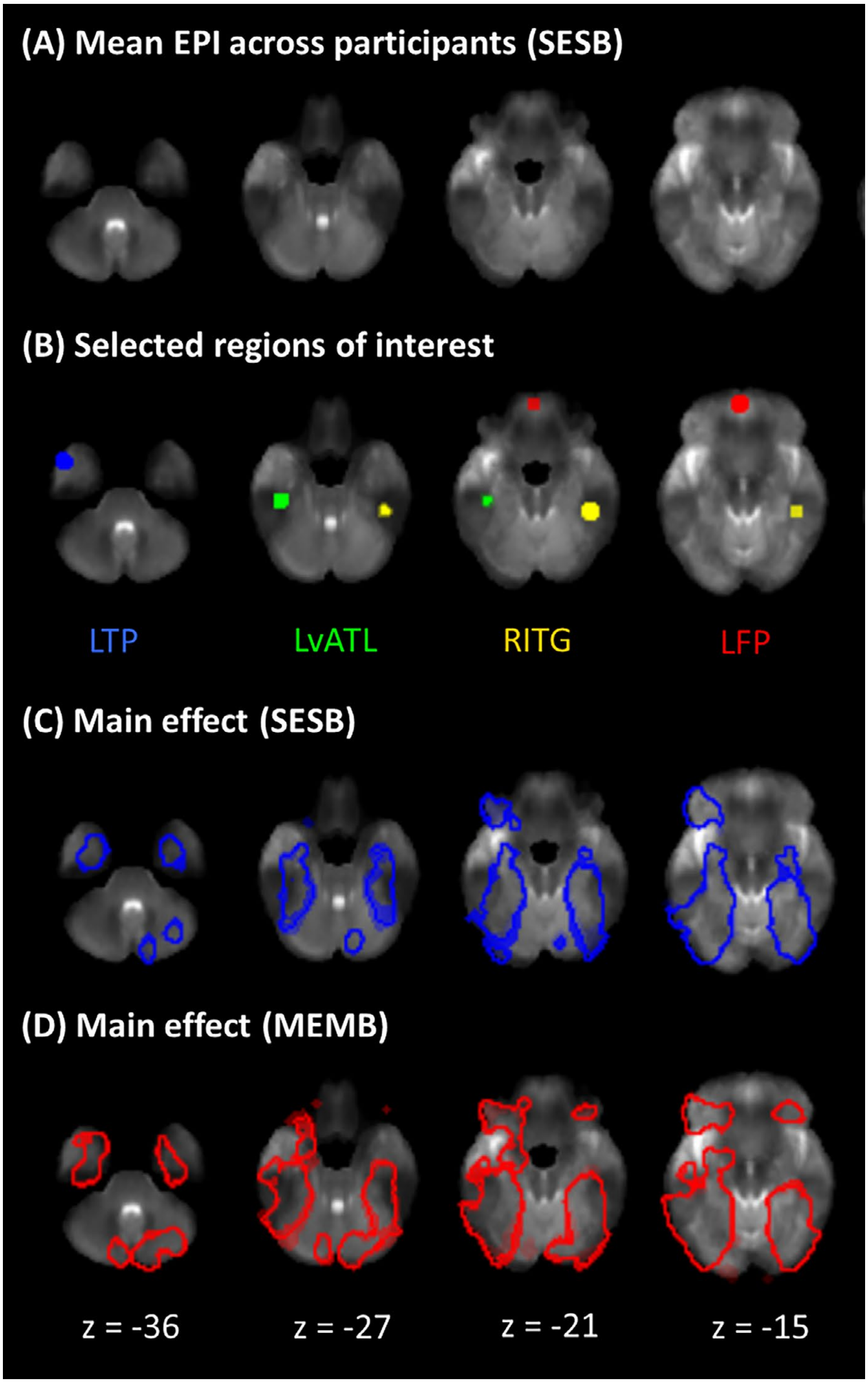

FIGURE 6 | Contrast despite magnetic field inhomogeneity. (A) Mean EPI across all participants for the baseline sequence (SESB) registered to MNI152NLin2009cAsym space. (B) Selected regions of interest overlaid on the mean EPI image. (C) Contour of the contrast of interest (S > C) for the baseline sequence. (D) Contour of the contrast of interest (S > C) for the MEMB sequence. Results are cluster-corrected at $p < 0.05$ based on an uncorrected voxel threshold of $p < 0.001$.

As hypothesised, MB improved activation precision in areas of inhomogeneity and in agreement with other work (Puckett et al. 2018). Figure S4 shows two very small clusters in which MB reduced activation precision, but which lie outside regions recruited by the task (Jung et al. 2017). After downsampling the MB timeseries to match the number of volumes in the SB timeseries, there was no longer any difference between the SB and MB sequences; this suggests that it is the increase in the number of volumes that accounts for the benefits of MB. Additionally, there was no evidence for signal leakage into the simultaneously-acquired slice location (McNabb et al. 2020; Todd et al. 2016), at least for our modest multiband factor of 2. We note that MB and ME seem to have independent effects as the interaction was not significant within any ROI (Table S1). However, ME seems to offer larger activation magnitude (Figure 6) and MB offers better activation precision. MB can help reduce the longer TR associated with ME and may decrease noise aliasing in studies using short TRs (~1 s).

Although this study focused primarily on univariate effects, we also identified improvements to multivariate metrics that will be of interest to researchers using increasingly-popular MVPA tools (Frisby et al. 2023). ME was found to improve MVPA performance compared to SE sequences and, in turn, MEdn sequences performed significantly better than ME sequences without denoising in every region of interest (with no detrimental effects). Future studies employing multivariate methods should strongly consider using both ME acquisitions and ME-ICA preprocessing.

This study provides evidence supporting use of ME and/or MB sequences to detect signal in regions prone to susceptibility artefacts. It should be noted that due to experimenter error, the flip angles for the SE sequences were not set to the Ernst angle, which could in theory have accounted for the ME benefit that we observed. However, acquisition of SE data with the flip angle set to the Ernst angle indicated that, although use of a suboptimal flip angle compromised tSNR, there was no consistent effect on any of our univariate or multivariate effects of interest (methods and results of this investigation are described in detail in Supplementary Material 2). The absence of a difference between the flip-angle-optimised SE sequence and the SE sequence that was actually used indicates that the advantage of ME relative to SE is robust to this flip angle error, although future studies should directly verify that this advantage holds and should seek to replicate our results using the Ernst angle for all sequences.

It is also important to note that individual 7T MRI scanners have their own hardware and software limitations; this means that each site must optimise and test feasible parameters locally. For example, our system did not allow us to take advantage of the higher spatial resolution offered by 7T-fMRI while retaining an adequately short first echo for our ME sequences. Puckett et al. (2018) used comparable voxel sizes but had a shorter first TE than us (9.9 vs. our 11.8 ms); Miletić et al. (2020) had higher resolution (1.6 vs. our 2.5 mm isotropic) and a shorter first echo of 9.66 ms. While we could in principle have achieved a smaller voxel size for our SE sequences, this would have been accompanied either by a narrower FOV or, if we were to maintain the FOV by increasing the number of slices, by a longer TR (which we wished to keep constant across sequences). As a second example, our 7T Terra pTx system imposes very conservative SAR limits (1 W per channel, 8 W total) for pTx scans but permits 20 W total (i.e., 2.5 W per channel) for circularly polarised mode (CP-mode or 'TrueForm') scans (equivalent to single transmit). We therefore had to use VERSE-modified pTx pulses which are known to be more sensitive to B_0 inhomogeneity. Third, we also acknowledge that pTx can be combined with ME or MB (Ding et al. 2023; Wu et al. 2016). Although we used the pTx head coil for all scans, we were not able to create sequences that combined pTx with ME or MB because the software to support this was not implemented on our scanner at the time of running this study. Future studies should seek to test the potential benefits of these combinations. It is possible that studies seeking to test more extreme parameter manipulations (numbers of echoes, multiband factors, or pTx approaches) than we have employed may reach different conclusions to us. For example, Todd et al. (2016) demonstrated that, although multiband factors of 3 or greater can sometimes be associated with greater BOLD sensitivity than the multiband factor of 2 that we have used, higher multiband factors are also associated with a risk of slice leakage. Fourth and finally, model-based image reconstruction methods (Yarach et al. 2017) may represent a promising approach for mitigating susceptibility-related image distortions and improving image quality in affected regions. However, these approaches primarily address off-resonance distortions and blurring rather than recovering signal irreversibly lost through intravoxel dephasing. They also require access to raw *k*-space data and entail substantial data storage and computational demands, which may limit their feasibility at some sites. In summary, rather than identifying the limits of what is possible (as others have done), we aim to offer an accessible framework that will enable researchers to leverage 7T-fMRI for investigating the functional roles of regions affected by susceptibility artefacts despite the hardware and software constraints of individual scanners.

5 | Conclusion

In this study we compared pTx, ME and MB as methods of improving sensitivity in temporal and frontal areas prone to magnetic susceptibility artefacts. Both pTx and ME improved activation magnitude, but only ME showed improvements in artefact-prone regions. MB and ME-ICA improved activation precision in these areas. Exploratory results suggested that both ME and ME-ICA may also benefit MVPA. We demonstrated that a multi-echo, multiband sequence running on the 8Tx32Rx pTx head coil (in CP mode) can detect signal in susceptible regions while maintaining sensitivity across the whole brain and is therefore a versatile choice for future studies using high field strength and investigating the functional roles of ventral temporal and/or orbitofrontal cortex (Binney et al. 2016; Borghesani et al. 2016; Devereux et al. 2018; Duncan 2010; DuPre et al. 2016; Fernandez et al. 2017; Lambon Ralph et al. 2017; Zahn et al. 2007).

Author Contributions

Saskia L. Frisby: conceptualisation, methodology, investigation, formal analysis, writing – original draft, writing – review and editing, visualisation. **Marta M. Correia:** conceptualisation, methodology, writing–review and editing, funding acquisition. **Minghao Zhang:** methodology, writing–review and editing. **Christopher T. Rodgers:** methodology, writing–review and editing, supervision. **Timothy T. Rogers:** writing–review and editing, supervision. **Matthew A. Lambon Ralph:** writing–review and editing, supervision, funding acquisition. **Ajay D. Halai:** conceptualisation, methodology, formal analysis, writing–review and editing, supervision, funding acquisition.

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Conflicts of Interest

C.T.R. receives research support from Siemens.

Data Availability Statement

Data are publicly available for scientific research purposes via the MRC Cognition and Brain Sciences Unit Data Repository (https://www.mrc-cbu.cam.ac.uk/publications/opendata/request/9273/). Code is publicly available at: https://github.com/slfrisby/7TOptimisation/.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** p values for all statistical tests within regions of interest (ROIs) based on the semantic network. $*p<0.05$, $**p<0.05$ (Bonferroni-corrected for the number of ROIs), — = ROI does not overlap by at least one voxel with the whole-brain contrast of interest (S > C) summed over all sequences in the comparison. LFP = left frontal pole; LIFGpt = left inferior temporal gyrus pars triangularis; LmMTG = left medial middle temporal gyrus; LpMTG = left posterior middle temporal gyrus; LTP = left temporal pole; LvATL = left ventral anterior temporal lobe; RITG = right inferior temporal gyrus **Table S2:** Significant cluster and peak information for main effects, ANOVA effects of interest and directed effects of interest when comparing activation magnitude and activation precision. All coordinates are given in MNI space and all anatomical labels are extracted from the Harvard-Oxford cortical and subcortical structural atlases. **Figure S1:** Effects of parallel transmit on activation magnitude. (A) effects of the SESB sequence and the pTx sequence for the contrast of interest (S > C); (B) effect of parallel transmit (pTx > SESB, red; SESB > pTx, no significant clusters). Results are cluster-corrected at $p<0.05$ based on an uncorrected voxel threshold of $p<0.001$ and are overlaid on the MNI152NLin2009cAsym template. **Figure S2:** Effects of parallel transmit on activation precision. Effects of the SESB sequence and the pTx sequence for the contrast of interest (S > C). No contrasts (pTx > SESB, SESB > pTx) were significant. Results are cluster-corrected at $p<0.05$ based on an uncorrected voxel threshold of $p<0.001$ and are overlaid on the MNI152NLin2009cAsym template. **Figure S3:** Effects of multi-echo and multiband on activation magnitude. (A) main effect of each sequence for the contrast of interest (S > C); (B) effect of echo (ME > SE, red; SE > ME, no significant clusters). No effects of multiband (MB > SB; SB > MB) were significant. Results are cluster-corrected at $p<0.05$ based on an uncorrected voxel threshold of $p<0.001$ and are overlaid on the MNI152NLin2009cAsym template. **Figure S4:** Effects of multi-echo and multiband on activation precision. (A) main effect of each sequence (S > C); (B) effect of echo (ME > SE, red; SE > ME, blue); (C) effect of band (MB > SB, red; SB > MB, blue). Results are cluster-corrected at $p<0.05$ based on an uncorrected voxel threshold of $p<0.001$ and are overlaid on the MNI152NLin2009cAsym template. **Figure S5:** Effect of ME-ICA denoising on activation magnitude (MEdn > ME, red; ME > MEdn, no significant clusters). Results are cluster-corrected at $p<0.05$ based on an uncorrected voxel threshold of $p<0.001$ and are overlaid on the MNI152NLin2009cAsym template. **Figure S6:** Effect of ME-ICA denoising on activation precision (MEdn > ME, red; ME > MEdn, blue). Results are cluster-corrected at $p<0.05$ based on an uncorrected voxel threshold of $p<0.001$ and are overlaid on the MNI152NLin2009cAsym template. **Figure S7:** Slice leakage analysis (all peaks). (A) Seed and possible artefact locations for a single participant. The ROIs (radius four voxels) indicate the seed location (blue), possible artefact location based on phase shift (green), possible artefact location based on GRAPPA (red), and possible artefact location based on phase shift and GRAPPA (yellow) in native EPI space. Informed consent was obtained from the participant for this image to be published. (B) Mean activation magnitude within each sphere for each MB sequence and the corresponding control sequence for each seed location. (C) Mean MVPA dissimilarity within each sphere for each MB sequence and the corresponding control sequence. **Table S3:** hbm70621-sup-0001-Supinfo.docx. p values for all slice leakage tests. A is the seed location, B is the possible artefact location based on phase shift, Ag is the possible artefact location in the same slice as A based on GRAPPA, and Bg is the possible artefact location in the same slice as B based on GRAPPA. $*p<0.05$, $**p<0.05$ (Bonferroni-corrected for the number of peaks and possible artefact regions). SEMB = single-echo multiband; MEMB = multi-echo multiband. **Table S4:** Impact of suboptimal flip angles on tSNR and effects of interest. Values are percentage improvement using the optimised sequence relative to the sequence reported in the main text. LFP = left frontal pole; LIFGpt = left inferior temporal gyrus pars triangularis; LmMTG = left medial middle temporal gyrus; LpMTG = left posterior middle temporal gyrus; LTP = left temporal pole; LvATL = left ventral anterior temporal lobe; RITG = right inferior temporal gyrus; SESB = single-echo single band sequence reported in main text; SESBernst = single-echo single band sequence using Ernst angle as nominal flip angle; SEMB = single-echo multiband sequence reported in main text; SEMBernst = single-echo multiband sequence using Ernst angle as nominal flip angle.